

Best Programming Practice

1. All values as variables including Fixed, User Inputs, and Results
2. Proper naming conventions for all variables

```
String name = "Eric";  
double height = input.nextDouble();  
double totalDistance = distanceFromToVia + distanceViaToFinalCity;
```

3. Proper Program Name and Class Name
4. Follow proper indentation
5. Give comments for every step or logical block like a variable declaration or conditional and loop blocks

1. **Sample Program 1** - Create a program to check if 3 values are internal angles of a triangle.

IMP => Follow Good Programming Practice demonstrated below in all Practice Programs

Hint =>

- a. Get integer input for 3 variables named x, y, and z.
- b. Find the sum of x, y, and z.
- c. If the sum is equal to 180, print "The given angles are internal angles of a triangle" else print They are not

Java

```
// Creating Class with name TriangleChecker indicating the purpose is to  
// check if the internal angles add to 180  
import java.util.Scanner;  
  
class TriangleChecker {  
    public static void main(String[] args) {  
        // Create a Scanner Object  
        Scanner input = new Scanner(System.in);  
  
        // Get 3 input values for angles  
        int x = input.nextInt();  
        int y = input.nextInt();  
        int z = input.nextInt();  
  
        // Find the sum of all angles  
        int sumOfAngles = x + y + z;  
  
        // Check if sum is equal to 180 and print either true or false  
        System.out.println("The given angles " +x+ ", " +y+ ", " + z +  
            " add to " + sumOfAngles);
```

```

if (sumOfAngles == 180) {
    System.out.println("The given angles are internal angles of a " +
        "Triangle");
} else {
    System.out.println("The given angles are not internal angles " +
        "of a Triangle");
}

// Closing the Scanner Stream
input.close();
}
}

```

2. **Sample Program 2** - Create a program to find the sum of all the digits of a number given by a user.

Hint =>

- Get an integer input for the number variable.
- Create an integer variable sum with an initial value of 0.
- Create a while loop to access each digit of the number.
- Inside the loop, add each digit of the number to the sum.
- Finally, print the sum outside the loop

Java

```

// Create SumOfDigit Class to compute the sum of all digits of a number
import java.util.Scanner;

class SumOfDigits {

    public static void main(String[] args) {
        // Create a Scanner Object
        Scanner input = new Scanner(System.in);

        // Get input value for number
        int origNumber = input.nextInt();

        // Define variable number and sum initialized to zero
        int number = origNumber;
        int sum = 0;
    }
}

```

```
// Run while loop to access each digit of number
while (number != 0) {
    // Use number % 10 to find each digit of number from last
    int digit = number % 10;

    // add each digit to sum
    sum += digit;

    // Remove last digit from number essentially get the quotient
    number = number / 10;
}

// Print the sum and close the Scanner Stream
System.out.println("The sum of digit of number:" +origNumber+ " = " +
                    sum);
input.close();
}
}
```

Level 1 Practice Programs

1. Write a program to check if a number is divisible by 5

I/P => number

O/P => Is the number ____ divisible by 5? ____

```
import java.util.Scanner;
public class DivisibleByFive{
    public static void main(String []args){

        //creating the scanner object
        Scanner sc = new Scanner(System.in);

        //creating variable n to indicate the number entered by user
        int n;

        //taking the number as user input
        System.out.print("Enter a number : ");
        n = sc.nextInt();

        //checking whether the number is divisible by 5 or not
        if(n%5 == 0){
            System.out.print(" Is the number " + n + " divisible by
5? Yes ");
        } else{
            System.out.print(" Is the number " + n + " divisible by
5? No ");
        }

        //closing the scanner object
        sc.close();
    }
}
```

2. Write a program to check if the first is the smallest of the 3 numbers.

I/P => number1, number2, number3

O/P => Is the first number the smallest? ____

```
import java.util.Scanner;
public class SmallestNumberCheck {
    public static void main(String[] args) {

        // Create a Scanner object to take user input
        Scanner input = new Scanner(System.in);

        // Take user input for the first number
        System.out.print("Enter the first number (number1): ");
        int number1 = input.nextInt();

        // Take user input for the second number
        System.out.print("Enter the second number (number2): ");
        int number2 = input.nextInt();

        // Take user input for the third number
        System.out.print("Enter the third number (number3): ");
        int number3 = input.nextInt();

        // Check if the first number is the smallest
        boolean isFirstSmallest = (number1 < number2) && (number1 <
        number3);

        // Print the result
        System.out.println("Is the first number the smallest? " +
        isFirstSmallest);

        // Close the scanner
        input.close();
    }
}
```

- Write a program to check if the first, second, or third number is the largest of the three.

I/P => number1, number2, number3

O/P =>

Is the first number the largest? ____

Is the second number the largest? ____

Is the third number the largest? ____

```
import java.util.Scanner;
public class LargestNumberCheck {
```

```
public static void main(String[] args) {

    // Create a Scanner object to take user input
    Scanner input = new Scanner(System.in);

    // Take user input for the first number
    System.out.print("Enter the first number: ");
    int number1 = input.nextInt();

    // Take user input for the second number
    System.out.print("Enter the second number: ");
    int number2 = input.nextInt();

    // Take user input for the third number
    System.out.print("Enter the third number: ");
    int number3 = input.nextInt();

    // Check if each number is the largest
    boolean isFirstLargest = (number1 >= number2) && (number1
    >=number3);
    boolean isSecondLargest = (number2 >= number1) && (number2
    >=number3);
    boolean isThirdLargest = (number3 >= number1) && (number3
    >=number2);

    // Print the results
    System.out.println("Is the first number the largest? "
    +isFirstLargest);
    System.out.println("Is the second number the largest? "
    +isSecondLargest);
    System.out.println("Is the third number the largest? "
    +isThirdLargest);

    // Close the scanner
    input.close();
}
}
```

4. Write a program to check for the natural number and write the sum of n natural numbers

Hint =>

- A Natural Number is a positive integer (1,2,3, etc) sometimes with the inclusion of 0
- A sum of n natural numbers is $n * (n+1) / 2$

I/P => number

O/P => If the number is a positive integer then the output is

The sum of ____ natural numbers is ____

Otherwise

The number ____ is not a natural number

```
import java.util.Scanner;
public class NaturalNumberSum {
    public static void main(String[] args) {

        // Create a Scanner object to take user input
        Scanner input = new Scanner(System.in);

        // Take user input for the number
        System.out.print("Enter a number: ");
        int number = input.nextInt();

        // Check if the number is a natural number
        if (number >= 0) {

            // Calculate the sum of the first N natural numbers
            int sum = number * (number + 1) / 2;

            // Print the result
            System.out.println("The sum of " + number + " natural
number is " + sum);
        }
        else {

            // Print that the number is not a natural number
            System.out.println("The number " + number + " is not a
natural number");
        }

        // Close the scanner
        input.close();
    }
}
```

5. Write a program to check whether a person can vote, depending on whether his/her age is greater than or equal to 18.

Hint =>

- a. Get integer input from the user and store it in the age variable.
- b. If the person is 18 or older, print "The person can vote." Otherwise, print "The person cannot vote."

I/P => age

O/P => If the person's age is greater or equal to 18 then the output is

The person's age is ____ and can vote.

Otherwise

The person's age is ____ and cannot vote.

```
import java.util.Scanner;
public class VoteEligibility {
    public static void main(String[] args) {

        // Create a Scanner object to take user input
        Scanner input = new Scanner(System.in);

        // Take user input for the age
        System.out.print("Enter the person's age: ");
        int age = input.nextInt();

        // Check if the person's age is greater or equal to 18
        if (age >= 18) {
            System.out.println("The person's age is " + age + " and
can vote.");
        }
        else {
            System.out.println("The person's age is " + age + " and
cannot vote.");
        }

        // Close the scanner
        input.close();
    }
}
```


6. Write a program to check whether a number is positive, negative, or zero.

Hint =>

- Get integer input from the user and store it in the number variable.
- If the number is positive, print positive.
- If the number is negative, print negative.
- If the number is zero, print zero.

```
import java.util.Scanner;
public class NatureOfNumber{
    public static void main(String[] args) {

        // Create a Scanner object to take user input
        Scanner input = new Scanner(System.in);

        // Take user input for the number
        System.out.print("Enter a number: ");
        int number = input.nextInt();

        // Check if the number is positive, negative, or zero
        if (number > 0) {
            System.out.println("The number is positive.");
        }
        else if (number < 0) {
            System.out.println("The number is negative.");
        }
        else {
            System.out.println("The number is zero.");
        }

        // Close the scanner
        input.close();
    }
}
```

7. Write a program SpringSeason that takes two int values month and day from the command line and prints "Its a Spring Season" otherwise prints "Not a Spring Season".

Hint =>

- Spring Season is from March 20 to June 20

```
import java.util.Scanner;
public class SpringSeason {
```

```
public static void main(String[] args) {

    // Create a Scanner object to take user input
    Scanner input = new Scanner(System.in);

    // Take user input for the month
    System.out.print("Enter the month (1-12): ");
    int month = input.nextInt();

    // Take user input for the day
    System.out.print("Enter the day (1-31): ");
    int day = input.nextInt();

    // Check if the date falls within the Spring Season
    if ((month == 3 && day >= 20) || (month == 6 && day <= 20) ||
(month > 3 && month < 6)) {
        System.out.println("It's a Spring Season");
    }
    else {
        System.out.println("Not a Spring Season");
    }

    // Close the scanner
    input.close();
}
}
```

8. Write a program to count down the number from the user input value to 1 using a **while** loop for a rocket launch

Hint =>

- Create a variable counter to take user inputted value for the countdown.
- Use the **while** loop to check if the counter is 1
- Inside a **while** loop, print the value of the counter and decrement the counter.

```
import java.util.Scanner;
public class RocketLaunchCountdown {
    public static void main(String[] args) {

        // Create a Scanner object to take user input
        Scanner input = new Scanner(System.in);
```

```
// Take user input for the countdown start value
System.out.print("Enter the countdown start value: ");
int counter = input.nextInt();

// Perform the countdown using a while loop
while (counter >= 1) {
    // Print the value of the counter
    System.out.println(counter);
    // Decrement the counter
    counter--;
}

// Close the scanner
input.close();
}
```

9. Rewrite program 8 to do the countdown using the **for**-loop

```
import java.util.Scanner;
public class RocketLaunchCountdown2 {
    public static void main(String[] args) {

        // Create a Scanner object to take user input
        Scanner input = new Scanner(System.in);

        // Take user input for the countdown start value
        System.out.print("Enter the countdown start value: ");
        int counter = input.nextInt();

        // Perform the countdown using a for-loop
        for (int i = counter; i >= 1; i--) {
            // Print the value of the counter
            System.out.println(i);
        }

        // Close the scanner
        input.close();
    }
}
```

```
}
}
}
```

10. Write a program to find the sum of numbers until the user enters 0

Hint =>

- Create a variable total of type double initialize to 0.0. Also, create a variable to store the double value the user enters
- Use the **while** loop to check if the user entered is 0
- If the user entered value is not 0 then inside the while block add user entered value to the total and ask the user to input again
- The loop will continue till the user enters zero and outside the loop display the total value

```
import java.util.Scanner;
public class SumUntilZero2 {
    public static void main(String[] args) {

        // Create a Scanner object to take user input
        Scanner input = new Scanner(System.in);

        // Initialize the total variable
        double total = 0.0;

        // Variable to store the user-entered value
        double userInput;

        // Enter the first input
        System.out.print("Enter a number (enter 0 to stop): ");
        userInput = input.nextDouble();

        // Loop to continue until the user enters 0
        while (userInput != 0) {

            // Add the user input to the total
            total += userInput;

            // The user enter the next input
            System.out.print("Enter a number (enter 0 to stop): ");
            userInput = input.nextDouble();

        }
    }
}
```

```

        // Print the total value
        System.out.println("The total sum is " + total);

        // Close the scanner
        input.close();
    }
}

```

11. Rewrite the program 10 to find the sum until the user enters 0 or a negative number using **while** loop and break statement

Hint =>

- Use infinite while loop as in **while** (true)
- Take the user entry and check if the user entered 0 or a negative number to break the loop using **break**;

```

import java.util.Scanner;
public class SumUntilZeroNegative {
    public static void main(String[] args) {

        // Create a Scanner object to take user input
        Scanner input = new Scanner(System.in);

        // Initialize the total variable
        double total = 0.0;

        // Infinite while loop
        while (true) {

            // Take user input
            System.out.print("Enter a number (enter 0 or a negative number to stop): ");
            double userInput = input.nextDouble();

            // Check if the user entered 0 or a negative number
            if (userInput <= 0) {
                break;
            }

            // Add the user input to the total
            total += userInput;
        }

        // Print the total value
        System.out.println("The total sum is " + total);
    }
}

```

```
// Close the scanner
input.close();
}
}
```

12. Write a program to find the sum of n natural numbers using **while** loop compare the result with the formulae $n*(n+1)/2$ and show the result from both computations was correct.

Hint =>

- Take the user input number and check whether it's a Natural number
- If it's a natural number Compute using formulae as well as compute using **while** loop
- Compare the two results and print the result

```
import java.util.Scanner;
public class NaturalNumberSumComparison {
    public static void main(String[] args) {

        // Create a Scanner object to take user input
        Scanner input = new Scanner(System.in);

        // Take user input for the number
        System.out.print("Enter a natural number: ");
        int number = input.nextInt();

        // Check if the number is a natural number (greater than 0)
        if (number > 0) {

            // Calculate the sum using the formula
            int sumFormula = number * (number + 1) / 2;

            // Calculate the sum using a while loop
            int sumWhileLoop = 0;
            int i = 1;
            while (i <= number) {
                sumWhileLoop += i;
                i++;
            }

            // Print the results
            System.out.println("The sum of " + number + " natural
```

```

numbers using the formula is " + sumFormula);
        System.out.println("The sum of " + number + " natural
numbers using the while loop is " + sumWhileLoop);

        // Compare the results
        if (sumFormula == sumWhileLoop) {
            System.out.println("Both computations are
correct.");
        }
        else {
            System.out.println("There is a difference between
the two computations.");
        }
    }
    else {
        System.out.println("The number " + number + " is not a
natural number.");
    }
    // Close the scanner
    input.close();
}
}

```

13. Rewrite the program number 12 with the **for** loop instead of a while loop to find the sum of n Natural Numbers.

Hint =>

- Take the user input number and check whether it's a Natural number
- If it's a natural number Compute using formulae as well as compute using **for** loop
- Compare the two results and print the result

```

import java.util.Scanner;
public class SumOfNaturalNumberUsingForLoop{
    public static void main(String []args){

        Scanner sc = new Scanner(System.in);

        /*creating variables n, sumByFormula and sumByForLoop
indicating the number of
        natural number and sum of n natural numbers by formula and

```

```

for loop respectively*/
int n, sumByFormula, sumByForLoop = 0;

//taking user input
System.out.print("Enter a number :");
n = sc.nextInt();

//checking input validity
if(n>0)
{
    System.out.println("It is natural number.");
    //calculating sum by formula
    sumByFormula = n * (n + 1)/2;

    //calculating sum by for loop
    for(int i=1;i<=n;i++)
    {
        sumByForLoop += i;
    }

    //comparing both answers
    if(sumByFormula == sumByForLoop)
    {
        System.out.println("Both values matched and sum is
" + sumByForLoop);
    }
    else
    {
        System.out.println("Both values are mismatched \n
sum by formula is " + sumByFormula + " and sum by for loop is " +
sumByForLoop );
    }
}
else
{
    System.out.println("It is not a natural number.");
}

}
}

```


14. Write a Program to find the factorial of an integer entered by the user.

Hint =>

- For example, the factorial of 4 is $1 * 2 * 3 * 4$ which is 24.
- Take an integer input from the user and assign it to the variable. Check the user has entered a positive integer.
- Using a **while** loop, compute the factorial.
- Print the factorial at the end.

```
import java.util.Scanner;
public class Factorial{
    public static void main(String []args){

        Scanner sc = new Scanner(System.in);

        /*creating variables n and fact indicating the number and
        factorial and respectively*/
        int n, fact = 1;

        //taking user input
        System.out.print("Enter a number :");
        n = sc.nextInt();

        //checking input validity
        if(n>=0)
        {
            System.out.println("It is positive number.");

            //calculating factorial
            if(n == 0){
                fact = 1;
            }
            else{
                int i = 1;
                while(i<=n){
                    fact=fact*i;
                    i++;
                }
            }

            //printing the factorial
```

```

        System.out.println("factorial of " + n + " is " + fact);
    }
    else
    {
        System.out.println("It is negative number.");
    }
}
}

```

15. Rewrite program 14 using for loop

Hint =>

- Take the integer input, check for natural number and determine the factorial using for loop and finally print the result.

```

import java.util.Scanner;
public class FactorialByForLoop{
    public static void main(String []args){

        Scanner sc = new Scanner(System.in);

        /*creating variables n and fact indicating the number and
        factorial and respectively*/
        int n, fact = 1;

        //taking user input
        System.out.print("Enter a number :");
        n = sc.nextInt();

        //checking input validity
        if(n>=0)
        {
            System.out.println("It is positive number.");

            //calculating factorial
            if(n == 0){
                fact = 1;
            }
            else{
                for(int i=1;i<=n;i++){

```

```

        fact *= i;
    }
}

//printing the factorial
System.out.println("factorial of " + n + " is " + fact);

}
else
{
    System.out.println("It is negative number.");
}
}
}

```

16. Create a program to print odd and even numbers between 1 to the number entered by the user.

Hint =>

- Get an integer input from the user, assign to a variable number and check for Natural Number
- Using a for loop, iterate from 1 to the number
- In each iteration of the loop, print the number is odd or even number

```

import java.util.Scanner;
public class EvenOrOdd{
    public static void main(String []args){

        Scanner sc = new Scanner(System.in);

        /*creating variables n and fact indicating the number and
        factorial and respectively*/
        int n, fact = 1;

        //taking user input
        System.out.print("Enter a number :");
        n = sc.nextInt();

        //checking input validity
        if(n>0)
        {

```

```

        System.out.println("It is natural number.");

        //iterating from 1 to n and printing whether number is
odd or even
        for(int i=1;i<=n;i++){

            if(i%2 == 0)
            {
                System.out.println(i + " is even number.");
            }
            else{
                System.out.println(i + " is odd number.");
            }
        }

    }
    else
    {
        System.out.println("It is negative number.");
        //iterating from -1 to n and printing whether number is
odd or even
        for(int i = -1;i>=n;i--){

            if(i%2 == 0)
            {
                System.out.println(i + " is even number.");
            }
            else{
                System.out.println(i + " is odd number.");
            }
        }

    }

}
}
}

```

17. Create a program to find the bonus of employees based on their years of service.

Hint =>

- a. Zara decided to give a bonus of 5% to employees whose year of service is more than 5 years.

- b. Take salary and year of service in the year as input.
- c. Print the bonus amount.

```
import java.util.Scanner;
public class Bonus{
    public static void main(String []args){

        Scanner sc = new Scanner(System.in);

        /*creating variables salary, yearOfService and bonus to
        indicate
        salary of employee, year of service and bonus amount
        respectively*/
        double salary, yearOfService, bonus = 0;

        //taking salary and year of service as user input
        System.out.print("Enter your salary :");
        salary = sc.nextInt();

        System.out.print("Enter year of service :");
        yearOfService = sc.nextInt();

        if(yearOfService > 5){
            bonus += salary * 5.0/100.0;
        }

        //printing the bonus amount
        System.out.println("Your bonus amount is " + bonus);
    }
}
```

18. Create a program to find the multiplication table of a number entered by the user from 6 to 9.

Hint =>

- a. Take integer input and store it in the variable number
- b. Using a for loop, find the multiplication table of number from 6 to 9 and print it in the format number * i = ____

```
import java.util.Scanner;
public class NumberTable{
    public static void main(String []args){
```

```
Scanner sc = new Scanner(System.in);

/*creating variable n indicating the number*/
int n;

//taking user input
System.out.print("Enter a number :");
n = sc.nextInt();

for(int i=6;i<=9;i++)
{
    System.out.println(n + " * " + i + " = " + n*i );
}
}
```